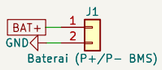
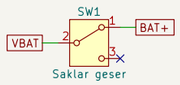
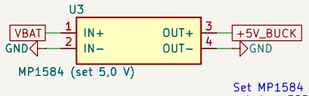
## 1. DAYA (wiring diagram blok 1)

J1 = P+/P− BMS (baterai 2S di rangka)




SW1: kaki tengah (B) = VBAT

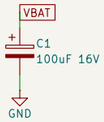



Set MP1584 ke 5,0 V dulu sebelum ESP32 dipasang

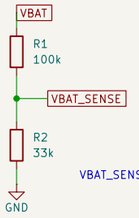
D1: cegah arus balik saat USB dicolok



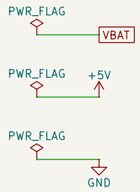

PWR_FLAG: daya VBAT, +5V, GND datang dari konektor/dioda



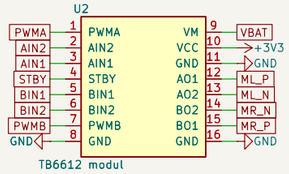

C1 dekat pin VM TB6612

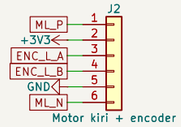


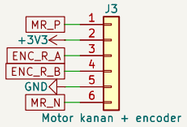

VBAT_SENSE ke GPIO39 (ADC1)




## 2. DRIVER MOTOR + MOTOR (wiring diagram blok 2)




Urutan pin modul TB6612 beda tiap penjual: cocokkan dengan modul yang dibeli.

VM = VBAT (7,4 V) langsung, VCC = 3,3 V logika. AO ke motor kiri, BO ke motor kanan.

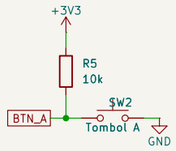
J2/J3 = JST PH 6 pin. Encoder tanpa resistor (firmware INPUT_PULLUP).

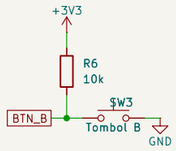
Urutan 6 kabel motor N20 beda tiap penjual: cek sebelum crimp.

## 3. LAYAR, TOMBOL, BUZZER (wiring diagram blok 3)

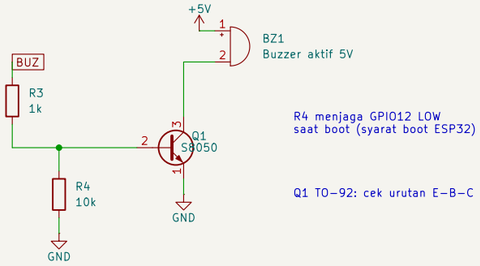



Cek urutan pin modul OLED (ada yang VCC dulu baru GND)



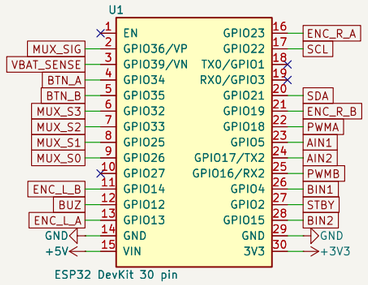

R5/R6 wajib: GPIO34/35 tanpa pull-up internal




R4 menjaga GPIO12 LOW saat boot (syarat boot ESP32)

Q1 TO-92: cek urutan E-B-C

## 4. ESP32 DEVKIT (otak)



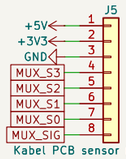

GPIO34/35/36/39: hanya input

GPIO2 (STBY) dan GPIO12 (BUZ) harus LOW saat boot

EN, GPIO27, TX0, RX0: tidak dipakai (X)

Footprint W25.4: ukur jarak baris pin modul (22,86 mm -> W22.86)

## 5. KABEL KE PCB SENSOR (wiring diagram blok 4)




J5 = JST XH 8 pin, urutan pin SAMA dengan J1 PCB sensor (kabel lurus, pin n ke pin n)

Ikuti NOMOR pin: pin 4–7 terbalik dari urutan nama (S3 dulu)

Di PCB: J5 sejajar tepi kanan hidung, pin 1 di ujung belakang

Dibuat oleh hardware/tools/gen_sch_utama.py, pin ESP32 sesuai CLAUDE.md
Panduan per blok: docs/wiring-diagram.html
**P01 Kel.5 PjBL DTEO FV ITS**

| | | |
|---|---|---|
| Sheet: / | | |
| File: pcb-utama.kicad_sch | | |
| ***Title: LTM – PCB utama*** | | |
| Size: A3 | Date: 2026-10-22 | **Rev: A** |
| KiCad E.D.A. 10.0.6 | | Id: 1/1 |